Research Project – Log Book

IT22556370 · Jayaweera M.P.Y.C · Style and Colour Recognition · 20.03.2026 – 18.10.2026

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Project ID	R26-IT-054 → AI-Powered Vision System for Real time Garment Tracking in Apparel Manufacturing.
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Name	Jayaweera M.P.Y.C
Individual Component	Cloth Pieces Counting and Operator Cycle Time Analysis.

Summary of the Individual Component

During this month, I worked on understanding my Style and Colour Recognition feature more clearly after the proposal presentation. A major contribution was the factory visit to MAS Sleekline, Nittambuwa, where I observed how finished garments are passed to the quality checker after several workers complete the sewing process. From that, I identified that the most suitable place for my feature is near the quality checker, using a camera to support automatic style and colour inspection. I also studied my main objective and specific objectives, including image capture, preprocessing, HSV-based colour detection, colour classification, comparison with expected production specifications, and future evaluation using performance metrics.

Date	RP Work
20 - 03 - 2026	• We had our proposal presentation on this day. • After the presentation, I understood my feature more clearly. • I noted the feedback given by the panel and identified the areas I need to improve in my Style and Colour Recognition component.

Date	RP Work
21 - 03 - 2026 to 22 - 03 - 2026	• I reviewed my proposal document carefully. • I focused on understanding the role of Style and Colour Recognition in the full system. • I understood that my feature is important for identifying garment style and garment colour automatically. • I studied the background of my feature. • I understood why manual checking of style and colour can create errors in apparel production. • This helped me understand the real problem my feature is trying to solve.
23 - 03 - 2026	• The research objectives related to this component were studied. • It was identified that this work should support garment style recognition and colour identification in the quality checking process. • Simple notes were also prepared for better understanding.
24 - 03 - 2026 to 25 - 03 - 2026	• The way this feature fits into the full research project was studied. • It was understood that the feature is connected with production monitoring and quality analysis. • The output is expected to support the system dashboard and decision process. • Attention was also given to garment style recognition. • The way the system can identify different garment types using visible features was studied. • The main garment classes related to the feature were written down.
26 - 03 - 2026	• Attention was given to colour recognition. • The way colour can be identified from garment images was studied. • It became clear that correct colour recognition is important for quality checking.

Date	RP Work
27 - 03 - 2026	• A factory visit was made to observe how the real working process happens at MAS Sleekline, Nittambuwa. • This visit helped to understand the actual garment production flow better. • It gave a practical idea about where this feature can be applied in the real factory environment. • It was also observed that after sewing is completed by the workers, the finished garment is handed over to the quality checker. • This showed that the quality checker is the most suitable point for this feature. • The quality checker checks garment quality while also identifying its style and colour. • This observation directly matched the assigned research feature. • From this understanding, it was identified that a camera should be installed near the quality checker to capture the garment for inspection. • This appears to be the practical way to implement the feature in the real environment. • This also helped show that style and colour checking should happen at the correct final stage.
28 - 03 - 2026 to 29 - 03 - 2026	• After the factory visit, the observations were reviewed carefully. • The actual production flow and the role of the quality checker became more clear. • This helped confirm the correct point where style and colour recognition should be applied in the system. • The idea of using a camera near the inspector became an important part of the feature plan.

Date	RP Work
30 - 03 - 2026 to 31 - 03 - 2026	• The work of the quality checker with the finished garment was studied again in relation to the research feature. • It was understood that style and colour checking happens at the final garment inspection stage. • This directly matched the feature responsibility in the quality checking process. • This stage was identified as the best place for capturing garment images for recognition.
01 - 04 - 2026	• The main objective was reviewed again after understanding the real factory process. • It was confirmed that the main goal is to develop a computer vision based system to automatically identify garment colours and colour patterns in real time to improve quality control accuracy.
02 - 04 - 2026 to 03 - 04 - 2026	• Factory observations were connected with the feature objectives. • It became clear that the feature should not only detect colour, but also support the quality checking process in a practical and real-time way. • The first specific objective of the feature was also studied. • It was understood that an image acquisition module is needed to capture garment images using a camera. • After the factory visit, this objective became clearer because the camera placement was already understood.

Date	RP Work
04 - 04 - 2026 to 06 - 04 - 2026	• The way colour detection should be done in this feature was studied. • It was understood that HSV colour space analysis is a suitable method to identify colours present in garments and extract dominant colour features. • This gave a stronger technical understanding of the colour detection part. • The colour detection logic was also studied further. • It was understood that the system should identify the main visible colours of the garment from the captured image.
07 - 04 - 2026 to 08 - 04 - 2026	• The colour classification objective was studied. • It was understood that the system should classify whether garments contain a single colour or multiple colour combinations such as black and white. • Thought was also given to how this classification can help the quality checker. • If the system can automatically identify the colour combination, it can reduce manual checking effort. • This made the practical value of the feature more clear.
09 - 04 - 2026	• The comparison stage of the feature was studied. • It was understood that detected garment colours should be compared with predefined production specifications to identify colour mismatches or colour changes during manufacturing.

Date	RP Work
10 - 04 - 2026 to 11 - 04 - 2026	• The system flow was reviewed again based on this feature. • It was understood that first the image must be captured, then preprocessed, then colours must be detected and classified. • After that, the results should be compared with expected production values. • The role of the decision-making part of the system was also studied. • It was understood that if the detected colour result does not match the expected garment colour, the system should identify it as a mismatch. • This is important for supporting real-time quality control.
13 - 04 - 2026 to 14 - 04 - 2026	• The way this feature fits into the overall system architecture was reviewed. • It was understood that the feature should connect with the monitoring and alerting process. • This means the recognition result should not stop at classification, but must support the full inspection workflow. • Practical challenges that may affect the feature were also studied. • It was understood that lighting conditions, folds, wrinkles and garment position can affect style and colour recognition accuracy. • These were noted as important things to consider in future implementation.
15 - 04 - 2026	• The way feature performance can be measured was reviewed. • It was understood that the system should later be evaluated using accuracy, precision, recall and F1-score. • This helped explain how the success of the component will be checked.

Date	RP Work
16 - 04 - 2026 to 17 - 04 - 2026	• The feature work was organized into clear parts. • It was separated into image capture, preprocessing, colour detection, colour classification, comparison and evaluation. • This gave a better and more structured direction for the future.
18 - 04 - 2026	• This month's contribution was summarized. • The main work during this month was understanding the actual quality checking process, identifying the correct point to apply the feature, and studying the technical steps needed for implementation. • This created a strong base for future development work.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	20th April – 2nd May
Planned Completion of My Component	November

Date	RP Work
20 - 04 - 2026	• The monthly progress report was finalized for submission.
22 - 04 - 2026 to 23 - 04 - 2026	• More attention was given to the image capture part of the feature. • It was understood that the camera position near to the quality checker must be planned carefully to get a clear garment image. • This is important because correct image capture affects all later steps such as preprocessing and colour detection.
25 - 04 - 2026	• The preprocessing stage was reviewed again in more detail. • This helped strengthen the technical understanding of how to prepare images for better recognition results.
27 - 04 - 2026 to 28 - 04 - 2026	• The colour detection logic was studied again with more focus on the practical side. • It became more clear that HSV colour space analysis is suitable for identifying visible garment colours under production conditions. • This was important for improving understanding of how the system can detect dominant and included colours from the garment image.

Date	RP Work
29 - 04 - 2026 to 30 - 04 - 2026	• Attention was given to the colour classification stage. • It was understood that the feature should identify whether the garment has a single colour or multiple colour combinations. • This is useful for the quality checker because the system can support faster and more accurate checking of garment colour patterns.
01 - 05 - 2026	• The comparison stage was also reviewed again, where detected colours must be checked against expected production specifications.
02 - 05 - 2026	• The overall feature flow was reviewed once more. • At this point, the process became clear as: image capture, preprocessing, colour detection, colour classification, comparison with expected values and support for the monitoring process.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	2nd May – 16th May
Planned Completion of My Component	November

Date	RP Work
02 - 05 - 2026	• The feature direction was reviewed again in relation to the full project. • At this stage, the role of Style and Colour Recognition became more clear within the garment tracking and quality checking process. • This helped strengthen the understanding of how this component should support the final system.
04 - 05 - 2026 to 05 - 05 - 2026	• More attention was given to the image acquisition stage. • It was understood that the camera should capture the finished garment clearly near the quality checker for better inspection results. • This is important because the image capture stage affects preprocessing, colour detection and classification accuracy.
07 - 05 - 2026	• The preprocessing stage was reviewed again in more detail. • It was understood that resizing, noise reduction, brightness adjustment and normalization are important to improve image quality before colour analysis. • This helped improve the technical understanding of the early steps of the feature.

Date	RP Work
09 - 05 - 2026 to 10 - 05 - 2026	• Attention was given to the colour detection logic. • It became more clear that HSV colour space analysis is suitable for identifying garment colours because it helps separate colour information from brightness. • This is useful when garments are inspected under real production conditions. • The importance of extracting dominant colour features from garment images was also understood more clearly.
12 - 05 - 2026	• The colour classification stage was reviewed again. • It was understood that garments should be classified according to single-colour and multi-colour combinations. • This is important because some garments may contain more than one visible colour, and the system should identify the correct colour combination during inspection.
14 - 05 - 2026	• The comparison stage of the feature was studied further. • It was understood that the detected garment colours should be checked against predefined production specifications. • If the detected result does not match the expected colour pattern, the system should identify it as a mismatch. • This is important for improving quality control accuracy in apparel manufacturing.
16 - 05 - 2026	• The full feature flow was reviewed once again. • At this point, the process became more structured as: image capture, preprocessing, HSV-based colour detection, colour classification, comparison with expected values, and support for the quality checking workflow.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	17th May – 31st May
Planned Completion of My Component	November

Date	RP Work
17 - 05 - 2026	• The feature flow was reviewed again to make sure each stage is clearly understood. Attention was given to how image capture, preprocessing, colour detection, colour classification and comparison should work together in one process.
19 - 05 - 2026 to 20 - 05 - 2026	• More attention was given to the image quality side of the feature. • It was understood that the captured garment image should be clear enough for correct colour analysis. This is important because poor image quality can affect both colour detection and style recognition results. This helped strengthen the understanding of why preprocessing is necessary before analysis.
22 - 05 - 2026	• The colour detection stage was studied again in a more practical way. It became more clear that the system should identify the major visible colours of the garment from the inspection image. This is important because the next classification step depends on correct colour extraction.

Date	RP Work
24 - 05 - 2026 to 25 - 05 - 2026	• The classification stage was reviewed again. It was understood that the system should identify whether the garment contains a single colour or a combination of colours. • This is useful for the quality checker because garments may contain simple colours or mixed colour patterns. This helped make the expected output clear.
27 - 05 - 2026	• Attention was given to the comparison stage once more. • It was understood that the detected colours should be checked against the expected production specification. If the detected result is different from the expected result, the garment should be marked as a mismatch. This is important for supporting quality control in the production process.
29 - 05 - 2026	• The style recognition side of the feature was also reviewed again. • It was understood that together with colour recognition, garment style should also be identified correctly during inspection. • This helped keep the feature aligned with the quality checker's actual work.
31 - 05 - 2026	• The overall progress of the feature was reviewed for this period.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	1st June – 30th June
Planned Completion of My Component	November

Date	RP Work
01 - 06 - 2026 to 04 - 06 - 2026	• Collection of garment images for the four style classes was continued. • Images were collected for SHIRT, T-SHIRT, TROUSER and SHORT. • Attention was given to keeping the number of images balanced across the four classes. • Annotation of the collected images was also started using Roboflow.
05 - 06 - 2026 to 09 - 06 - 2026	• The earlier frame-state model was reviewed again. • It was understood that the READY state appears only for a short moment in each packing cycle, so there were not enough samples of it for the model to learn properly. • Because of this, the trained frame-state model was removed from the working pipeline. • The method and the reason for removing it were written down so that the work is kept as a documented result.

Date	RP Work
08 - 06 - 2026 to 12 - 06 - 2026	• A rule based capture gate was designed to replace the removed model. • The order of the checks was decided as region of interest, area check, confidence, blur and consensus. • Every constant that can be tuned was written down together with the reason for it. • It was understood that this approach can be checked and adjusted without training a model again.
11 - 06 - 2026 to 15 - 06 - 2026	• The region of interest crop was implemented. • The folding zone was fixed between 0.15 and 0.85 of the frame width and 0.20 and 0.95 of the frame height. • The crop was checked against footage recorded from each factory. • It was understood that this gives the model the same kind of framing it was trained on.
15 - 06 - 2026 to 18 - 06 - 2026	• The blur rejection step was implemented using the variance of the Laplacian. • The threshold was set at 8.0 after testing on recorded video. • It was understood that this removes frames that are blurred while the garment is still moving in the operator's hands.
17 - 06 - 2026 to 20 - 06 - 2026	• The box area plausibility check was implemented. • Boxes smaller than 15% or larger than 85% of the region of interest were rejected. • It was understood that this protects against a box that covers the whole frame instead of one garment.

Date	RP Work
19 - 06 - 2026 to 24 - 06 - 2026	• The rolling window consensus was implemented. • Two good frames within the last four attempts are required before a capture is made. • It was understood that this is better than requiring two frames in a row, because one blurred frame does not remove all the progress for that garment. • The dataset version was also generated with a 70/15/15 train, validation and test split and exported in YOLO format.
22 - 06 - 2026 to 26 - 06 - 2026	• The table clear and auto unlock handling was implemented. • Five clear frames release the lock, and an automatic unlock happens after ninety frames so the line does not stop. • The training configuration for the garment style detection model was also prepared. • YOLOv8n was selected because the system has to run on a factory computer without a GPU.
23 - 06 - 2026 to 29 - 06 - 2026	• The garment style detection model was trained for 50 epochs. • Training was done on a CPU only laptop and took about 10 hours. • The trained weights were saved as best.pt. • The training results and the plots were archived for the report.
26 - 06 - 2026 to 01 - 07 - 2026	• The capture thresholds were tuned against the recorded factory videos. • The captures made by the system were compared with a manual count. • The constants were adjusted and the test was run again until the behaviour was acceptable.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	1st July – 31st July
Planned Completion of My Component	November

Date	RP Work
01 - 07 - 2026 to 03 - 07 - 2026	• The predictions of the garment style detection model were evaluated. • The training curves were plotted and the checkpoints were compared. • It was noted that the best result was mAP@0.5 of 0.987 at epoch 41. • The requirements for the monitoring application were also defined.
04 - 07 - 2026 to 07 - 07 - 2026	• The Electron and React application shell was set up with Vite and TypeScript. • The inference speed was benchmarked at image sizes 640 and 384. • It was found that 640 takes about 51 ms per frame and 384 about 27 ms per frame. • 384 was selected for deployment because packing is fast and a higher frame rate is more useful than a small accuracy gain. • The requirements for colour recognition were also defined.

Date	RP Work
08 - 07 - 2026 to 10 - 07 - 2026	• The style detection model was integrated with the camera pipeline. • The detection output was passed through the confidence and area checks. • Foreground segmentation of the garment was implemented using GrabCut on an inset crop. • It was understood that this separates the garment from the table before any colour is measured.
11 - 07 - 2026 to 14 - 07 - 2026	• The FastAPI backend service was built. • The database schema for garment records was designed and created in SQLite. • Colour clustering was implemented using k-means in the CIELAB colour space. • CIELAB was selected instead of HSV because distance in that space matches how colour difference is seen by the eye, and there is no wrap-around problem at red.
15 - 07 - 2026 to 17 - 07 - 2026	• A reference palette of fifteen colours was built. • Each cluster centre is given a name by finding the nearest colour in the palette. • It was understood that this replaces the earlier hand written hue bands with a cleaner method.
18 - 07 - 2026 to 20 - 07 - 2026	• The cluster retention threshold was tuned and set at 1.5% of the foreground pixels. • Clusters below this share are treated as shadow, print detail or noise. • The camera pipeline was connected to the backend so that every capture is posted and stored with a timestamp.

Date	RP Work
21 - 07 - 2026 to 24 - 07 - 2026	• The colour names produced by the system were checked against physical garments. • Any disagreements were written down and the palette entries were adjusted where needed.
25 - 07 - 2026 to 27 - 07 - 2026	• Colour recognition was integrated into the camera pipeline. • The live monitoring dashboard was built, showing the capture feed, running counts by style and the colour of each capture.
28 - 07 - 2026 to 31 - 07 - 2026	• The validation was re-run on the held out test split. • The requirements for the production analytics part were defined. • The introduction and problem statement chapter of the thesis was drafted.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	1st August – 31st August
Planned Completion of My Component	November

Date	RP Work
01 - 08 - 2026 to 03 - 08 - 2026	• The effective working hours calculation was implemented. • Effective hours are the scheduled shift minus the fixed breaks minus any logged downtime. • Downtime is subtracted using interval overlap, so a breakdown that crosses midnight is counted correctly on both days. • The order management screens were also built and the field evaluation requirements were defined.
04 - 08 - 2026 to 06 - 08 - 2026	• The packing rate per effective hour and the three day recent average were implemented. • Days that were completely lost to downtime are skipped instead of being counted as zero. • The literature review chapter was drafted. • The camera was installed and aligned above the folding zone at the packing station.
07 - 08 - 2026 to 09 - 08 - 2026	• End to end capture and storage was validated against the live camera. • The system was run across a full production shift for the first time. • The order progress and efficiency summary was implemented, showing target against packed and the required rate against the current rate.

Date	RP Work
10 - 08 - 2026 to 11 - 08 - 2026	• The projected completion date and the delay in days were implemented. • A false detection was investigated where a person's face and hair were classified as a SHIRT at 75.49% confidence. • It was understood that this happens because the model has only four classes and no class for 'not a garment', so it always returns the best of the four.
12 - 08 - 2026 to 13 - 08 - 2026	• The next day output forecast was implemented using a five day weighted moving average blended with a linear trend. • The confidence threshold for capture was raised from 0.75 to 0.85 and the trial was run again. • This value was chosen because every genuine capture observed was at 82% or above, while the false positive was between 75% and 78%.
14 - 08 - 2026 to 15 - 08 - 2026	• The full frame box collapse problem was documented. • It was found that almost every training box covers the whole image, so on an uncropped wide shot the model boxes the entire frame. • This explains why the benchmark score is higher than the real field behaviour, and why the region of interest crop is necessary. • The methodology chapter was drafted and the analytics dashboard views were built.
17 - 08 - 2026 to 21 - 08 - 2026	• The analytics output was validated by recomputing one past week by hand and comparing it with the dashboard. • The preparation plan for Progress Presentation 2 was defined.

Date	RP Work
24 - 08 - 2026 to 26 - 08 - 2026	• The results and evaluation chapter was drafted. • The presentation speech and the slide deck were written. • A question and answer bank was built covering the model, the two deployment failures, colour, the database, the backend and the analytics. • The known weak points were listed with an honest answer prepared for each one.
27 - 08 - 2026 to 31 - 08 - 2026	• The submission requirements for the ICAC 2026 conference were confirmed. • The conference paper was written in the IEEE double column template within the six page limit. • The figures cheat sheet was memorised and the presentation was rehearsed against the clock. • The live system demonstration was prepared, with a recorded clip kept as a fallback.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	1st September – 30th September
Planned Completion of My Component	November

Date	RP Work
01 - 09 - 2026 to 03 - 09 - 2026	• The conference paper was submitted through Microsoft CMT on the closing date. • Progress Presentation 2 was delivered to the panel and the feedback was written down. • The submitted paper and the submission confirmation were archived.
04 - 09 - 2026 to 08 - 09 - 2026	• The field trial records were extracted by querying the database directly on the factory machine. • The record count, the date range and the distribution by style and colour were exported as a summary. • The weight ordering in the next day forecast was corrected, because the weights were being applied to the oldest day instead of the newest. • The confusion matrix was rebuilt at the confidence threshold that is actually deployed.
09 - 09 - 2026 to 12 - 09 - 2026	• The requirements for the project website were defined. • The corrected test split metrics were written into the documentation, replacing the earlier validation figures. • The discussion and limitations chapter was drafted.

Date	RP Work
14 - 09 - 2026 to 16 - 09 - 2026	• It was recorded as a stated limitation that no manual counting baseline was measured, so no claim of efficiency gain is made. • The structure and content of the website were planned. • The end to end capture precision and miss rate were measured against a manual count and written into the results chapter.
17 - 09 - 2026 to 19 - 09 - 2026	• A technical reference set was compiled covering the methodology, the models and algorithms, and the calculations. • Every figure was checked against the source code so that nothing is written from memory. • The figures, screenshots and demonstration media for the website were prepared. • The references, figures and appendices of the thesis were compiled.
21 - 09 - 2026 to 25 - 09 - 2026	• The research overview and methodology pages of the website were written. • The thesis was finalised after supervisor feedback, with each chapter revised and the formatting and figure numbering checked.
28 - 09 - 2026 to 30 - 09 - 2026	• The results and findings pages were added to the website. • The same honest framing used in the thesis was kept on the website, that a high benchmark score is not by itself evidence of reliability in the field.

Project	AI-Powered Vision System for Real-Time Garment Tracking in Apparel Manufacturing.
Feature	Style and Colour Recognition.
Period Covered	1st October – 18th October
Planned Completion of My Component	November

Date	RP Work
01 - 10 - 2026 to 05 - 10 - 2026	• The downtime and rework logging screens were built, so that operators can record breakdowns and power failures from the application. • The documents, downloads and contact pages were added to the website.
06 - 10 - 2026 to 08 - 10 - 2026	• The website was deployed and every internal and external link was checked. • The site was tested on both mobile and desktop and the media was confirmed to load.
09 - 10 - 2026 to 11 - 10 - 2026	• The draft thesis was finalised, the plagiarism check was run and supervisor approval was obtained. • The draft thesis was submitted on the faculty deadline. • The project website was submitted for evaluation.
12 - 10 - 2026 to 14 - 10 - 2026	• The MS Planner plan was completed and exported, and the Teams workspace was organised by component. • The final check list was completed with the supporting evidence and submitted. • A plan and timeline for the camera-ready version of the conference paper was prepared in advance.

Date	RP Work
15 - 10 - 2026 to 17 - 10 - 2026	• The presentation was updated for the final viva. • The question and answer bank was refreshed so that it carries the corrected test split metrics and the field evaluation figures instead of the earlier validation numbers.
18 - 10 - 2026	• The final live demonstration was prepared and a full dress rehearsal was completed. • The log book and the RP evidence pack were assembled for submission.